Radiative transitions of strange heavy axial mesons in light-cone QCD sum rules

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Abstract

We calculate the radiative widths of $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ in light-cone QCD sum rules. The two basis-current sum rules are combined into the physical axial-vector amplitudes using the same mixing convention as in the hadronic residues. With the lattice-normalized leading-twist photon input, we obtain

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 32.7^{+6.7}_{-6.2}\ \text{keV}, \qquad \Gamma[D_{s1}(2536) \to D_s\gamma] = 0.425^{+1.352}_{-0.390}\ \text{keV}.$$

The smallness of the $D_{s1}(2536) \to D_s\gamma$ width is traced to destructive interference between the two physical-current components. In the bottom-strange sector we find

$$\Gamma[B_{s1}(5750) \to B_s\gamma] = 19.3^{+7.9}_{-5.2}\ \text{keV}, \qquad \Gamma[B_{s1}(5830) \to B_s\gamma] = 0.504^{+1.173}_{-0.447}\ \text{keV},$$

where the lower state is treated as a theoretical diagnostic with $M_{B_{s1}(5750)} \simeq 5.75$ GeV. The results support a strong hierarchy between the radiative widths of the two axial states and provide predictions for future measurements of bottom-strange radiative transitions.

I. INTRODUCTION

Radiative transitions of heavy-light mesons provide a sensitive probe of their internal structure. In positive-parity strange heavy mesons, such transitions are especially useful because they depend on the spin composition of the physical axial-vector states and on the coupling of the photon to both the heavy and strange quark degrees of freedom. They therefore complement mass spectroscopy and strong-decay measurements in testing the assignment of the observed states and the role of spin mixing.

The charm-strange axial-vector mesons $D_{s1}(2460)$ and $D_{s1}(2536)$ are particularly important examples. The $D_{s1}(2460)$ is a very narrow state below the D^*K threshold and its radiative decay to $D_s\gamma$ has been observed in B decays and in continuum production [1?]. The $D_{s1}(2536)$, on the other hand, is well established in hadronic channels such as D^*K and $D\pi K$ [2, 3], but its radiative decay to $D_s\gamma$ has not been observed with comparable significance. The measured width of the $D_{s1}(2536)$ is below the MeV scale, which makes a suppressed radiative branching fraction experimentally difficult to isolate [3]. In the bottom-strange sector the $B_{s1}(5830)$ state has been observed in B^*K spectroscopy [4?], whereas no direct information is available on its radiative transition to $B_s\gamma$.

Theoretical studies of these radiative modes have been carried out using constituent quark models, relativistic quark models, effective approaches, and QCD sum rules [5–16]. The light-cone QCD sum-rule calculation of Ref. [7] showed that $D_{s1}(2460) \to D_s\gamma$ is expected to be one of the leading radiative channels of the $D_{s1}(2460)$. More recently, Bondar and Milstein argued that the smallness of $D_{s1}(2536) \to D_s\gamma$ follows from a cancellation between the two spin components of the physical axial-vector state [15, 16]. Their analysis also considered the corresponding bottom-strange transitions, including both the established $B_{s1}(5830)$ state and a lower $B_{s1}(5750)$ configuration. The spread among existing predictions is particularly large for the channels where the two axial-current components interfere destructively.

In this work we calculate the radiative transitions $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ in light-cone QCD sum rules. We keep the strange-quark-mass terms and use photon distribution amplitudes, including the recent lattice-normalized input for the leading-twist photon distribution amplitude. The two basis-current sum rules are combined into the physical axial-vector amplitudes, which allows us to test whether the cancellation mechanism proposed for $D_{s1}(2536) \to D_s\gamma$ is reproduced in the sum-rule framework. For the bottom-strange sector, the lower $B_{s1}(5750)$ state is treated as a theoretical diagnostic, while the $B_{s1}(5830)$ result is a prediction for the experimentally established state.

The paper is organized as follows. In Sec. II we define the interpolating currents, the physical-current mixing, the radiative form factor, and the light-cone sum rules. Section III contains the numerical analysis, comparison with previous studies, and discussion of the experimental status. We summarize our results in Sec. IV.

II. THEORETICAL FRAMEWORK

We consider the radiative transition of an axial-vector strange heavy meson with momentum $p' = p+q$ into a pseudoscalar strange heavy meson with momentum p and a real photon with momentum q. The heavy quark is denoted by $Q = c, b$, while the light quark is $q = s$ in the applications considered below. The pseudoscalar interpolating current is chosen as

$$J_5 = \bar{q}\,\mathrm{i}\gamma_5 Q. \tag{1}$$

For the axial-vector channel we use two independent basis currents,

$$J_\mu^A = \bar{q}\gamma_\mu\gamma_5 Q, \tag{2}$$

$$J_\mu^B = \frac{\mathrm{i}}{m_Q + m_q}\,\bar{q}\sigma_{\mu\alpha}p'^\alpha\gamma_5 Q, \tag{3}$$

where p'^α is the momentum of the initial axial-vector meson. The physical-current combinations are written as

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \tag{4}$$

$$J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B. \tag{5}$$

With this convention, $J_{1\mu}$ is assigned to the lower axial-vector combination, whereas $J_{2\mu}$ is assigned to the higher one. Thus, in the charm-strange sector, J_1 is associated with $D_{s1}(2460)$ and J_2 with $D_{s1}(2536)$. In the bottom-strange sector, the same convention associates J_1 with the lower diagnostic configuration $B_{s1}(5750)$ and J_2 with the established $B_{s1}(5830)$. We use $\theta = 35.3^\circ$ as the central value and vary it in the interval $25^\circ \le \theta \le 45^\circ$ in the numerical analysis.

The decay constants are defined by

$$\langle 0|\bar{q}\,\mathrm{i}\gamma_5 Q|P(p)\rangle = \frac{m_P^2 f_P}{m_Q + m_q}, \tag{6}$$

$$\langle 0|J_{i\mu}|X_i(p',\eta)\rangle = f_i m_i \eta_\mu, \qquad i = 1,2, \tag{7}$$

where X_i denotes the axial-vector state coupled to $J_{i\mu}$, with mass m_i, polarization vector η_μ, and decay constant f_i.

The radiative matrix element is parametrized by a single gauge-invariant electric-dipole form factor,

$$\langle P(p)\gamma(q,\varepsilon)|X_i(p',\eta)\rangle = e g_i\left[(\eta\cdot\varepsilon^*)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon^*)\right]. \tag{8}$$

The corresponding partial width is

$$\Gamma(X_i \to P\gamma) = \frac{\alpha_{\rm em}}{3}\, g_i^2 \left(\frac{m_i^2 - m_P^2}{2m_i}\right)^3. \tag{9}$$

Following the three-point notation of Rohrwild [17], the light-cone sum rule starts from

$$\Pi_{i\mu\nu}(p,q) = \mathrm{i}^2 \int d^4x\, d^4y\, e^{\mathrm{i}p\cdot x + \mathrm{i}q\cdot y}\langle 0|T\{J_5(x) j_\nu^{\rm em}(y) J_{i\mu}^\dagger(0)\}|0\rangle, \qquad i = 1,2, \tag{10}$$

where

$$j_\nu^{\rm em} = e_q \bar{q}\gamma_\nu q + e_Q \bar{Q}\gamma_\nu Q \tag{11}$$

and the charges are in units of the positive elementary charge. Equivalently, introduce the plane-wave background field

$$F_{\alpha\beta}(z) = \mathrm{i}(q_\alpha \varepsilon^*_\beta - q_\beta \varepsilon^*_\alpha) e^{\mathrm{i}q\cdot z}. \tag{12}$$

The term linear in $F_{\alpha\beta}$ satisfies

$$\varepsilon^{*\nu}\Pi_{i\mu\nu}(p,q) = \mathrm{i}\int d^4x\, e^{\mathrm{i}p\cdot x}\langle 0|T\{J_5(x)J^\dagger_{i\mu}(0)\}|0\rangle_F. \tag{13}$$

The commonly used vacuum-to-photon correlator is precisely this linear background-field component, rather than an independent starting convention. Since $J_{1\mu}$ and $J_{2\mu}$ are linear combinations of J^A_μ and J^B_μ, we first calculate the two basis correlators and only afterwards form the physical combinations. We use the coefficient of the gauge-invariant structure

$$(p\cdot q)\varepsilon^*_\mu - (p\cdot\varepsilon^*)q_\mu. \tag{14}$$

For the basis currents we write

$$\Pi^K_\mu(p,q) = \Pi_K(p^2,p'^2)\left[(p\cdot q)\varepsilon^*_\mu - (p\cdot\varepsilon^*)q_\mu\right] + \cdots, \qquad K = A, B, \tag{15}$$

where the remaining Lorentz structures are not used in the sum rule.

On the hadronic side, isolating the ground-state double pole gives

$$\Pi^{\rm had}_i(p^2,p'^2) = \frac{m_P^2 f_P}{m_Q + m_q}\,\frac{m_i f_i g_i}{(m_i^2 - p'^2)(m_P^2 - p^2)} + \cdots, \qquad i = 1, 2, \tag{16}$$

where the ellipsis denotes higher-state and continuum contributions.

The QCD side is obtained from the light-cone expansion in the external electromagnetic field. For the heavy-quark line we use

$$\begin{aligned} S_Q(k) = \frac{\not{k} + m_Q}{m_Q^2 - k^2} &- e_Q\int_0^1 du\left[\frac{\not{k} + m_Q}{2(m_Q^2-k^2)^2}F^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu F^{\mu\nu}(ux)\gamma_\nu}{m_Q^2 - k^2}\right] \\ &- g_s\int_0^1 du\left[\frac{\not{k} + m_Q}{2(m_Q^2-k^2)^2}G^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu G^{\mu\nu}(ux)\gamma_\nu}{m_Q^2 - k^2}\right] + \cdots. \end{aligned} \tag{17}$$

The first term gives the free heavy-quark propagator. The terms proportional to e_Q describe perturbative photon emission from the heavy-quark line, whereas the terms proportional to

g_s generate the quark-gluon photon matrix elements. For the strange line we use Rohrwild's coordinate-space notation, but retain the mass terms omitted in the massless expression of Ref. [17]. With $\bar{u} = 1 - u$,

$$
\begin{aligned}
\widehat{s^a(x)\bar{s}^b}(0) = \delta^{ab}\left[\frac{\mathrm{i}\,\not{x}}{2\pi^2 x^4} - \frac{m_s}{4\pi^2 x^2}\right] \\
&- \frac{\mathrm{i}g_s}{16\pi^2 x^2}\int_0^1 du\,\{\bar{u}\,\not{x}\sigma_{\alpha\beta} + u\sigma_{\alpha\beta}\,\not{x}\}G_A^{\alpha\beta}(ux)t_{ab}^A \\
&- \frac{\mathrm{i}e_s\delta^{ab}}{16\pi^2 x^2}\int_0^1 du\,\{\bar{u}\,\not{x}\sigma_{\alpha\beta} + u\sigma_{\alpha\beta}\,\not{x}\}F^{\alpha\beta}(ux) \\
&- \frac{m_s}{32\pi^2}\int_0^1 du\,\left[g_s G_A^{\alpha\beta}(ux)t_{ab}^A + e_s\delta^{ab}F^{\alpha\beta}(ux)\right]\sigma_{\alpha\beta}L_x + \cdots\,,
\end{aligned}
\tag{18}
$$

where

$$
L_x = \ln\left(\frac{-x^2\mu^2}{4}\right) + 2\gamma_E. \tag{19}
$$

No local vacuum-condensate term is displayed in Eq. (18). Setting m_s and the last line to zero recovers Rohrwild's displayed light propagator. The nonlocal vacuum-to-photon quark and quark-gluon matrix elements define the two- and three-particle photon DAs collected in Appendix A.

We emphasize an implementation point relevant to the numerical tables: the existing axial-current code still includes the local heavy-line photon contribution named `heavy_local`, proportional to $e_Q e^{-m_Q^2/M^2}\langle\bar{s}s\rangle$. It is not part of the displayed propagator above and has not been silently removed from the numerical results. The mixed and vacuum-gluon condensate contributions were not included in the radiative three-point calculation.

After the double Borel transformation in p'^2 and p^2 and continuum subtraction, the basis-current invariant amplitudes are written as

$$
\widehat{\Pi}_K(M_1^2, M_2^2, s_0) = \widehat{\Pi}_K^{\mathrm{pert}} + \delta_{KA}\widehat{\Pi}_A^{\mathrm{local}} + \widehat{\Pi}_K^{(2p)} + \widehat{\Pi}_K^{(3p)}, \qquad K = A, B. \tag{20}
$$

Here the terms denote perturbative photon emission, the separately implemented local light-quark-condensate contribution in the present axial calculation, two-particle photon-DA contributions, and three-particle photon-DA contributions, respectively. No local condensate term is thereby being inserted into either propagator displayed above. At the symmetric Borel point used in the numerical analysis,

$$
M_1^2 = M_2^2 = 2M^2, \qquad u_0 = \frac{M_1^2}{M_1^2 + M_2^2} = \frac{1}{2}, \tag{21}
$$

the perturbative parts have the generic form

$$\widehat{\Pi}_K^{\rm pert} = \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2} \rho_K(s), \qquad K = A, B. \tag{22}$$

The photon-DA conventions and auxiliary combinations entering $\widehat{\Pi}_K^{(2p)}$ and $\widehat{\Pi}_K^{(3p)}$ are given in Appendix A.

The physical invariant amplitudes are obtained by applying the same rotation as in Eq. (5),

$$\widehat{\Pi}_1 = \sin\theta\, \widehat{\Pi}_A + \cos\theta\, \widehat{\Pi}_B, \tag{23}$$

$$\widehat{\Pi}_2 = \cos\theta\, \widehat{\Pi}_A - \sin\theta\, \widehat{\Pi}_B. \tag{24}$$

Matching the hadronic and QCD representations gives

$$g_i = \frac{m_Q + m_q}{m_P^2 f_P\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_P^2}{M_2^2}\right) \widehat{\Pi}_i(M_1^2, M_2^2, s_0), \qquad i = 1, 2. \tag{25}$$

Equivalently,

$$g_1 = \frac{m_Q + m_q}{m_P^2 f_P\, m_1 f_1} \exp\left(\frac{m_1^2}{M_1^2} + \frac{m_P^2}{M_2^2}\right) \left[\sin\theta\, \widehat{\Pi}_A + \cos\theta\, \widehat{\Pi}_B\right], \tag{26}$$

$$g_2 = \frac{m_Q + m_q}{m_P^2 f_P\, m_2 f_2} \exp\left(\frac{m_2^2}{M_1^2} + \frac{m_P^2}{M_2^2}\right) \left[\cos\theta\, \widehat{\Pi}_A - \sin\theta\, \widehat{\Pi}_B\right]. \tag{27}$$

The first expression is used for $D_{s1}(2460) \to D_s\gamma$ and $B_{s1}(5750) \to B_s\gamma$, while the second is used for $D_{s1}(2536) \to D_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$.

A. Physical-current two-point functions and decay constants

For clarity, the physical residues entering Eqs. (26) and (27) are associated with the transverse two-point matrix

$$\Pi_{\mu\nu}^{XY}(p') = \mathrm{i} \int d^4x\, e^{\mathrm{i}p'\cdot x} \langle 0 | T\{J_\mu^X(x) J_\nu^{Y\dagger}(0)\} | 0 \rangle, \qquad X, Y = A, B. \tag{28}$$

Suppressing the common transverse tensor, the same state rotation gives

$$\Pi_{11} = \sin^2\theta\, \Pi_{AA} + \cos^2\theta\, \Pi_{BB} + 2\sin\theta\cos\theta\, \Pi_{AB}, \tag{29}$$

$$\Pi_{22} = \cos^2\theta\, \Pi_{AA} + \sin^2\theta\, \Pi_{BB} - 2\sin\theta\cos\theta\, \Pi_{AB}, \tag{30}$$

$$\Pi_{12} = \sin\theta\cos\theta(\Pi_{AA} - \Pi_{BB}) + (\cos^2\theta - \sin^2\theta)\Pi_{AB}. \tag{31}$$

If

$$\widehat{\Pi}_{XY}=\int_{s_{\rm th}}^{s_0}ds\,e^{-s/\mathcal{M}^2}\rho^{\rm pert}_{XY}(s)+\sum_d\widehat{\Pi}^{(d)}_{XY}, \tag{32}$$

the explicit physical combination can be kept compact as

$$\begin{aligned}\widehat{\Pi}_{11}=&\int_{s_{\rm th}}^{s_0}ds\,e^{-s/\mathcal{M}^2}\left[\sin^2\theta\rho^{\rm pert}_{AA}+\cos^2\theta\rho^{\rm pert}_{BB}+2\sin\theta\cos\theta\rho^{\rm pert}_{AB}\right]\\&+\sum_d\left[\sin^2\theta\widehat{\Pi}^{(d)}_{AA}+\cos^2\theta\widehat{\Pi}^{(d)}_{BB}+2\sin\theta\cos\theta\widehat{\Pi}^{(d)}_{AB}\right],\end{aligned} \tag{33}$$

with $\widehat{\Pi}_{22}$ obtained by the replacements shown explicitly in Eq. (30). The corresponding direct sum rules would be

$$f_1^2=\frac{e^{m_1^2/\mathcal{M}^2}}{m_1^2}\widehat{\Pi}_{11},\qquad\qquad f_2^2=\frac{e^{m_2^2/\mathcal{M}^2}}{m_2^2}\widehat{\Pi}_{22}. \tag{34}$$

In the present calculation the AA two-point OPE is available, but the full BB and AB OPEs have not been independently completed. Therefore the numerical f_1, f_2 values quoted below are overlap-model normalizations, not independent predictions of Eqs. (32) and (34).

III. NUMERICAL ANALYSIS

The external inputs and benchmark decay constants used in the numerical analysis are collected in Table I. The quark masses and hadron masses are taken from the Particle Data Group [18]. The pseudoscalar decay constants are taken from the lattice averages summarized by FLAG [19]. For the photon distribution amplitudes we use the BBK/Rohrwild convention [17, 20]. We compare two normalizations of the leading-twist photon matrix element: the conventional susceptibility normalization, proportional to $\chi_s\langle\bar{s}s\rangle$, and the recent lattice normalization $f^{\perp}_{\gamma,s}$ of the leading-twist photon DA [21]. The latter is used for our preferred numerical predictions, while the susceptibility-normalized results are retained to facilitate comparison with earlier photon-LCSR analyses.

The physical-current residues are not independent external inputs. The reduced overlap model used in the numerical scan is

$$\begin{aligned}f_1^2&=\sin^2\theta f_A^2+\cos^2\theta f_B^2+2\sin\theta\cos\theta\rho_{AB}f_Af_B,\\f_2^2&=\cos^2\theta f_A^2+\sin^2\theta f_B^2-2\sin\theta\cos\theta\rho_{AB}f_Af_B.\end{aligned} \tag{35}$$

TABLE I. External inputs and benchmark decay constants used in the numerical analysis. Photon-DA normalizations are quoted at $\mu = 1$ GeV.

Parameter	Value
$m_c,\ m_b,\ m_s$	$1.27(2),\ 4.18(3),\ 0.093(11)$ GeV [18]
$M_{D_{s1}(2460)},\ M_{D_{s1}(2536)},\ m_{D_s}$	$2.4595,\ 2.53511,\ 1.96835$ GeV [18]
$M_{B_{s1}(5830)},\ m_{B_s}$	$5.82870,\ 5.36692$ GeV [4, 18]
$M_{B_{s1}(5750)}$	$5.750(26)$ GeV [?]
$f_{D_s},\ f_{B_s}$	$0.2499,\ 0.2303$ GeV [19]
$f_A(D_{s1})$	$0.245(17)$ GeV [22]
$f_A^{(B_s)},\ f_T^{(B_s)}$	$0.305(27),\ 0.285(27)$ GeV [14]
$\chi_s(1\ \text{GeV})$	$+3.15(30)\ \text{GeV}^{-2}$ [17]
$f^{\perp}_{\gamma,s}(1\ \text{GeV})$	$-55.1(1.9)$ MeV [21]

For the charm-strange sector we use $f_1 = 0.345(17)$ GeV as the working normalization anchor, determine ρ_{AB} from the first equation, and then obtain f_2. At the central angle the saved scans give $f_1 = 0.344[0.329, 0.363]$ GeV and $f_2 = 0.380[0.339, 0.423]$ GeV for the legacy photon input, and $f_1 = 0.345[0.330, 0.364]$ GeV and $f_2 = 0.379[0.338, 0.423]$ GeV for the lattice-normalized input. For the bottom-strange sector, the basis-current constants of Ref. [14] fix the basis normalization and ρ_{AB} is chosen by the closure condition $\Pi_{12} = 0$. For the $B_{s1}(5830)$ scan this gives $f_2 = 0.218[0.138, 0.292]$ GeV (legacy) and $0.203[0.121, 0.291]$ GeV (lattice). These are the numerical values actually found, with the model qualification stated below Eq. (34).

The Borel windows and continuum thresholds are summarized in Table II. The windows are chosen by the standard sum-rule criteria: the prediction should be stable against variations of the Borel parameter, the continuum contribution should not dominate the dispersion integral, and the higher-twist photon-DA terms should remain under control. Since the initial and final mesons in each channel are close in mass, we use the symmetric Borel choice

$$M_1^2 = M_2^2 = 2M^2, \qquad u_0 = \frac{1}{2}, \tag{36}$$

as in Sec. II.

The central mixing angle is taken as $\theta = 35.3^\circ$, corresponding to the heavy-quark-limit

TABLE II. Auxiliary parameters used in the numerical analysis.

Channel	M^2 window	s_0 window
$D_{s1}(2460), D_{s1}(2536) \to D_s\gamma$	3.0–4.5 GeV^2	7.5–8.5 GeV^2
$B_{s1}(5750) \to B_s\gamma$	10.0–14.0 GeV^2	36.5–40.5 GeV^2
$B_{s1}(5830) \to B_s\gamma$	10.0–14.0 GeV^2	38.0–41.0 GeV^2

FIG. 1. Representative Borel and continuum-threshold stability checks for the charm-strange and bottom-strange channels.

value for the mixing of the two axial-vector configurations. To estimate the sensitivity to deviations from this limit and to phenomenological convention choices, we also scan the interval $25^\circ \leq \theta \leq 45^\circ$. This scan is treated as a systematic variation rather than as a statistical error. The angle dependence is particularly important for the channels in which the two basis-current contributions interfere destructively.

Representative stability plots are shown in Fig. 1. In the chosen working windows the form factors show acceptable stability under variations of M^2 and s_0. The residual dependence on these auxiliary parameters is included in the quoted uncertainties.

The uncertainties are obtained by varying the inputs listed in Tables I and II. Dimensionful quantities with quoted errors are sampled within their stated uncertainties, while the Borel parameter and continuum threshold are varied over the working intervals in Table II. Where indicated, the mixing angle is also varied in the interval $25^\circ \leq \theta \leq 45^\circ$. For each sample, the physical form factor is computed from Eq. (25), and the corresponding width is obtained from Eq. (9). We quote medians and 16–84 percentile intervals. This convention naturally gives asymmetric uncertainties and is more appropriate than a Gaussian fit for the cancellation-sensitive channels, especially $D_{s1}(2536) \to D_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$.

The final form factors and widths at the central mixing angle are listed in Table III. The $f^{\perp}_{\gamma,s}$ rows are our preferred predictions. The $\chi_s\langle\bar{s}s\rangle$ rows are retained to show the size of the photon-normalization dependence and to allow comparison with earlier photon-LCSR calculations.

The $B_{s1}(5750)$ entry is a lower bottom-strange diagnostic configuration rather than an

TABLE III. Form factors and radiative widths obtained at the central mixing angle $\theta = 35.3°$. Here f_X denotes the physical-current residue of the initial axial-vector state entering Eq. (25). The quoted uncertainties correspond to the 16–84 percentile intervals.

Channel	Photon input	f_X (GeV)	g_X (GeV^{-1})	Γ (keV)
$D_{s1}(2460) \to D_s\gamma$	$\chi_s\langle\bar{s}s\rangle$	$0.347^{+0.017}_{-0.018}$	$-0.555^{+0.077}_{-0.075}$	$64.8^{+18.4}_{-16.8}$
$D_{s1}(2536) \to D_s\gamma$	$\chi_s\langle\bar{s}s\rangle$	$0.382^{+0.043}_{-0.043}$	$+0.0031^{+0.0558}_{-0.0667}$	$0.640^{+1.13}_{-0.589}$
$D_{s1}(2460) \to D_s\gamma$	$f^{\perp}_{\gamma,s}$	$0.345^{+0.016}_{-0.016}$	$-0.394^{+0.039}_{-0.039}$	$32.7^{+6.7}_{-6.2}$
$D_{s1}(2536) \to D_s\gamma$	$f^{\perp}_{\gamma,s}$	$0.380^{+0.041}_{-0.043}$	$+0.0329^{+0.0427}_{-0.0432}$	$0.425^{+1.352}_{-0.390}$
$B_{s1}(5750) \to B_s\gamma$	$\chi_s\langle\bar{s}s\rangle$	$0.415^{+0.056}_{-0.055}$	$+0.282^{+0.067}_{-0.051}$	$10.0^{+7.0}_{-3.8}$
$B_{s1}(5750) \to B_s\gamma$	$f^{\perp}_{\gamma,s}$	$0.416^{+0.048}_{-0.054}$	$+0.395^{+0.068}_{-0.044}$	$19.3^{+7.9}_{-5.2}$
$B_{s1}(5830) \to B_s\gamma$	$\chi_s\langle\bar{s}s\rangle$	$0.246^{+0.059}_{-0.075}$	$-0.0364^{+0.0418}_{-0.0634}$	$0.375^{+1.74}_{-0.341}$
$B_{s1}(5830) \to B_s\gamma$	$f^{\perp}_{\gamma,s}$	$0.247^{+0.063}_{-0.075}$	$+0.0105^{+0.0498}_{-0.0898}$	$0.504^{+1.173}_{-0.447}$

FIG. 2. Monte Carlo width distributions. Left: charm-strange channels with the $f^{\perp}_{\gamma,s}$ photon input. Right: bottom-strange physical-current normalization.

established experimental resonance, and should therefore be read as a prediction for the corresponding mixed-current state if it is realized in nature.

The comparison of the two photon normalizations shows that the leading-twist photon input is one of the dominant sources of uncertainty. In the conventional susceptibility-normalized scenario, the result depends on the product $\chi_s\langle\bar{s}s\rangle$, which carries the uncertainty of both the magnetic susceptibility and the strange-quark condensate. The lattice-normalized input $f^{\perp}_{\gamma,s}$ replaces this product by a directly normalized leading-twist photon matrix element and gives the preferred modern normalization used in the phenomenological discussion below.

The suppression pattern can be seen directly at the amplitude level. With the $f^{\perp}_{\gamma,s}$ input and after applying the mixing coefficients, the two basis-current contributions to the normalized $D_{s1}(2460) \to D_s\gamma$ amplitude are -0.0715 and -0.205; they add constructively. For $D_{s1}(2536) \to D_s\gamma$, the corresponding contributions are -0.0917 and $+0.132$, so the physical amplitude is a small difference. This destructive interference explains the strong suppression of $D_{s1}(2536) \to D_s\gamma$ and provides the LCSR realization of the cancellation mechanism discussed in Refs. [15, 16]. A similar cancellation-sensitive pattern appears for the established $B_{s1}(5830)$ state.

The distributions in Fig. 2 illustrate why percentile intervals are used. The $D_{s1}(2460)$

distribution is relatively compact, whereas the $D_{s1}(2536)$ and $B_{s1}(5830)$ distributions are distorted by cancellations near the amplitude level. In such cases a symmetric Gaussian uncertainty would obscure the physical origin of the error.

Using the measured total width of the $D_{s1}(2536)$, our preferred result corresponds to a branching fraction of order

$$\mathcal{B}[D_{s1}(2536) \to D_s\gamma] \sim 5 \times 10^{-4}. \tag{37}$$

This is consistent with the absence of a clear experimental signal for this mode and provides a natural explanation for why it is difficult to isolate. For $D_{s1}(2460)$, the measured radiative branching fraction together with our preferred partial width is compatible with a very narrow total-width scale.

The results also clarify why the predictions in the literature span a wide range. In cancellation-sensitive channels, small changes in the relative size or sign of the two physical-current components can lead to large changes in the partial width. Future measurements of $D_{s1}(2536) \to D_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$, or improved upper limits on these modes, would therefore provide direct tests of the spin-mixing structure of strange heavy axial-vector mesons.

IV. CONCLUSION

Radiative transitions of positive-parity strange heavy mesons provide a sensitive test of the spin structure of the physical axial-vector states. In this work we studied the decays $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ within light-cone QCD sum rules. The calculation was performed for two basis axial currents, and the resulting amplitudes were combined into the physical 1^+ states using the same mixing convention as in the hadronic residues. We kept the strange-quark-mass terms and compared the conventional susceptibility normalization of the photon distribution amplitude with the lattice-normalized leading-twist photon input.

With the lattice-normalized photon input, our preferred charm-strange results are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 32.7^{+6.7}_{-6.2}\ \text{keV}, \tag{38}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 0.425^{+1.352}_{-0.390}\ \text{keV}. \tag{39}$$

The two widths are therefore strongly hierarchical. The $D_{s1}(2460) \to D_s\gamma$ width is at the tens-of-keV level, whereas $D_{s1}(2536) \to D_s\gamma$ is suppressed to the sub-keV level. This suppression is traced to destructive interference between the two physical-current components of the $D_{s1}(2536)$ amplitude. In contrast, the corresponding components in the $D_{s1}(2460)$ amplitude add constructively. Thus the light-cone sum-rule calculation gives an independent realization of the cancellation mechanism discussed in Refs. [15, 16].

The result is consistent with the present experimental situation. The radiative decay $D_{s1}(2460) \to D_s\gamma$ has been observed, whereas no clear signal exists for $D_{s1}(2536) \to D_s\gamma$. Using the measured total width of the $D_{s1}(2536)$, our preferred partial width corresponds to a branching fraction of order 5×10^{-4}, providing a natural explanation for why this channel is difficult to isolate experimentally.

In the bottom-strange sector, the preferred prediction for the established $B_{s1}(5830)$ state is

$$\Gamma[B_{s1}(5830) \to B_s\gamma] = 0.504^{+1.173}_{-0.447}\ \text{keV}. \tag{40}$$

For the lower $B_{s1}(5750)$ diagnostic configuration we find a larger radiative width. Since direct radiative information on bottom-strange axial mesons is not yet available, these results provide benchmarks for future measurements.

The comparison with earlier theoretical work shows that these radiative transitions remain phenomenologically important. Existing predictions span a wide range, especially for cancellation-sensitive channels. Our analysis indicates that the hierarchy between the two axial-vector states is controlled not only by phase space, but also by the relative sign and size of the two mixed-current contributions. Future measurements of $D_{s1}(2536) \to D_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$, or improved upper limits on these modes, would therefore provide direct tests of the spin-mixing structure of strange heavy axial mesons and of the radiative-transition dynamics described here.

Appendix A: Photon distribution amplitudes and auxiliary functions

For the nonperturbative photon matrix elements we use the photon distribution amplitudes (DAs) in the notation of Rohrwild [17], which is based on the BBK classification [20]. We collect here the functions and auxiliary combinations entering the sum rules of Sec. II. The variable u denotes the quark momentum fraction, $\bar{u} = 1 - u$, and $\alpha_q + \alpha_{\bar{q}} + \alpha_g = 1$ for the three-particle DAs.

The leading-twist two-particle DA is taken in the asymptotic form

$$\phi_\gamma(u) = 6u\bar{u}. \tag{A1}$$

The twist-3 vector DA is

$$\psi^{(v)}(u) = -20u\bar{u}(2u-1) + \frac{15}{16}\left(\omega_\gamma^A - 3\omega_\gamma^V\right)u\bar{u}(2u-1)\left[7(2u-1)^2 - 3\right]. \tag{A2}$$

For the twist-4 two-particle DAs we use

$$\begin{aligned}\mathcal{A}(u) = 40u\bar{u}\left(3\kappa - \kappa^+ + 1\right) + 8\left(\zeta_2^+ - 3\zeta_2\right)\Big[u\bar{u}(2+13u\bar{u}) \\ + u^3(10-15u+6u^2)\ln u + \bar{u}^3(10-15\bar{u}+6\bar{u}^2)\ln\bar{u}\Big],\end{aligned} \tag{A3}$$

$$\mathcal{B}(u) = 40(1+3\kappa^+)\int_0^u d\alpha\,(u-\alpha)\left[-\frac{1}{2} + \frac{3}{2}(2\alpha-1)^2\right]. \tag{A4}$$

The twist-4 three-particle DAs used in the numerical analysis are

$$\begin{aligned}\mathcal{S}(\alpha_i) = 30\alpha_g^2\Big[(\kappa+\kappa^+)(1-\alpha_g) + (\zeta_1+\zeta_1^+)(1-\alpha_g)(1-2\alpha_g) \\ + \zeta_2\{3(\alpha_{\bar{q}}-\alpha_q)^2 - \alpha_g(1-\alpha_g)\}\Big],\end{aligned} \tag{A5}$$

$$\begin{aligned}\widetilde{\mathcal{S}}(\alpha_i) = -30\alpha_g^2\Big[(\kappa-\kappa^+)(1-\alpha_g) + (\zeta_1-\zeta_1^+)(1-\alpha_g)(1-2\alpha_g) \\ + \zeta_2\{3(\alpha_{\bar{q}}-\alpha_q)^2 - \alpha_g(1-\alpha_g)\}\Big],\end{aligned} \tag{A6}$$

$$\mathcal{T}_1(\alpha_i) = -120(3\zeta_2+\zeta_2^+)(\alpha_q-\alpha_{\bar{q}})\alpha_{\bar{q}}\alpha_q\alpha_g, \tag{A7}$$

$$\mathcal{T}_2(\alpha_i) = 30\alpha_g^2(\alpha_q-\alpha_{\bar{q}})\Big[\kappa-\kappa^+ + (\zeta_1-\zeta_1^+)(1-2\alpha_g) + \zeta_2(3-4\alpha_g)\Big], \tag{A8}$$

$$\mathcal{T}_3(\alpha_i) = -120(3\zeta_2-\zeta_2^+)(\alpha_q-\alpha_{\bar{q}})\alpha_{\bar{q}}\alpha_q\alpha_g, \tag{A9}$$

$$\mathcal{T}_4(\alpha_i) = 30\alpha_g^2(\alpha_q-\alpha_{\bar{q}})\Big[\kappa+\kappa^+ + (\zeta_1+\zeta_1^+)(1-2\alpha_g) + \zeta_2(3-4\alpha_g)\Big]. \tag{A10}$$

The two-particle functions entering the Borel-transformed amplitudes are used through

$$\Psi^{(v)}(u_0) = \int_0^{u_0} du\,\psi^{(v)}(u), \tag{A11}$$

$$\mathcal{B}(u_0) = -8\int_0^{u_0} du\,(u_0-u)h_\gamma(u), \tag{A12}$$

where Eq. (A12) is the notation dictionary connecting the Rohrwild/BBK convention to the form used in the single-current benchmark of Ref. [7].

TABLE IV. Photon-DA parameters used at $\mu = 1$ GeV in Rohrwild notation.

Parameter	Value	Comment
χ_s	3.15 ± 0.30 GeV^{-2}	legacy susceptibility input
$f_{3\gamma}$	$-(4.0 \pm 2.0) \times 10^{-3}$ GeV^2	twist-3 coupling
ω_γ^A	-2.1 ± 1.0	twist-3 shape
ω_γ^V	3.8 ± 1.8	twist-3 shape
κ	0.15	twist-4 shape
κ^+	-0.05	twist-4 shape
ζ_1	0.40	twist-4 shape
ζ_1^+	0	twist-4 shape
ζ_2	0.30	twist-4 shape
ζ_2^+	0	twist-4 shape

For the three-particle contribution we define

$$\mathcal{I}_F(u_0) = \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g \, F(u_0 - (1-v)\alpha_g,\, 1 - u_0 - v\alpha_g,\, \alpha_g,\, v) \\ + \int_{1-u_0}^{1} dv \int_0^{(1-u_0)/v} d\alpha_g \, F(u_0 - (1-v)\alpha_g,\, 1 - u_0 - v\alpha_g,\, \alpha_g,\, v)\,, \tag{A13}$$

with

$$F = \mathcal{S} + \widetilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v\,(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2)\,. \tag{A14}$$

All three-particle DAs in Eq. (A14) are evaluated at $(\alpha_q, \alpha_{\bar{q}}, \alpha_g)$, with the arguments specified in Eq. (A13). In the numerical analysis we use the symmetric point $u_0 = 1/2$.

In the preferred modern scenario we replace the legacy normalization $\chi_s\langle\bar{s}s\rangle$ of the leading-twist matrix element by the lattice-normalized input $f^{\perp}_{\gamma,s}$. The DA shapes in this appendix are otherwise kept fixed in the same Rohrwild/BBK convention.

Appendix B: Basis-current invariant amplitudes

We write the Borel-transformed invariant amplitudes in the form

$$\widehat{\Pi}_K(M^2, s_0) = \widehat{\Pi}_K^{\text{pert}} + \widehat{\Pi}_K^{(2p)} + \widehat{\Pi}_K^{(3p)}, \qquad K = A, B, \tag{B1}$$

where the three terms denote the perturbative photon emission, two-particle photon-DA contribution, and three-particle photon-DA contribution, respectively. At the symmetric Borel point used in the numerical analysis,

$$M_1^2 = M_2^2 = 2M^2, \qquad u_0 = \frac{1}{2},$$

the perturbative parts are written as

$$\widehat{\Pi}_K^{\text{pert}} = \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2} \rho_K(s), \qquad K = A, B. \tag{B2}$$

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